

DETERMINANTS AND MATRICES

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1. DETERMINANTS AND MATRICES

1.1 Basics

A. Definition

1.1: Matrix

A matrix is a rectangular array of numbers or quantities.

Let's look at an example of a matrix:

$$\begin{bmatrix} 4 & 7 & 9 \\ 2 & 3 & 10 \end{bmatrix}$$

The matrix above has six numbers arranged in:

- Three Columns
- Two Rows

In general, when naming matrices, we will use capital, boldface letters.

$$\mathbf{A} = \begin{bmatrix} 4 & 7 & 9 \\ 2 & 3 & 10 \end{bmatrix}$$

Rows are horizontal

$$\begin{aligned} \text{Row1} &= \begin{bmatrix} 4 & 7 & 9 \end{bmatrix} \\ \text{Row2} &= \begin{bmatrix} 2 & 3 & 10 \end{bmatrix} \end{aligned}$$

Columns are vertical

$$\text{Column 1} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \quad \begin{bmatrix} 7 \\ 3 \end{bmatrix} \quad \begin{bmatrix} 9 \\ 10 \end{bmatrix}$$

1.2: Order of a Matrix

- An $m \times n$ matrix has m rows and n columns
- $m \times n$ is called the order of the matrix

Example 1.3

State the order of the matrix

$$\begin{bmatrix} 4 & 7 & 9 \\ 2 & 3 & 10 \end{bmatrix}$$

The order of the matrix is:

$$\underbrace{2}_{\text{Rows}} \times \underbrace{3}_{\text{Columns}}$$

1.4: Elements of a Matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

- Each quantity at the intersection of a row and a column is an element.
- Elements are numbered $a_{\{\text{Row}\}\{\text{Column}\}}$

Example 1.5

Matrix **A** is a 2×3 matrix such that the value of the a^{th} element is the sum of its column number and its row number. Write **A**.

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

1.6: Square Matrix

In a matrix, if

$$m = n$$

That is, the number of rows is equal to the number of columns, then it is a square matrix.

$$X = \begin{bmatrix} 5 & 2 & 7 \\ 8 & 1 & 3 \\ 2 & 7 & 4 \end{bmatrix}, \quad X \text{ is a } 3 \times 3 \text{ Square Matrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad A \text{ is a } 2 \times 2 \text{ Square Matrix}$$

1.7: Diagonal of a Matrix

The diagonal of a matrix goes from top left to bottom right.

Example 1.8

- A. Identify the diagonal of the matrix $\begin{bmatrix} 5 & 2 & 7 \\ 8 & 1 & 3 \\ 18 & 7 & 4 \end{bmatrix}$. Identify the elements that lie on the diagonal.
- B. Identify the elements that lie on the diagonal of the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Part A

$$\begin{bmatrix} \mathbf{5} & 2 & 7 \\ 8 & \mathbf{1} & 3 \\ 18 & 7 & \mathbf{4} \end{bmatrix}$$

The elements **5**, **1** and **4** lie on the diagonal of the matrix.

Part B

$$\begin{bmatrix} \mathbf{1} & 2 \\ 3 & \mathbf{4} \end{bmatrix}$$

1.9: Types of Matrices

- In a matrix, if all the elements except those on the diagonal are zero, then the matrix is called a diagonal matrix. The term is usually applicable to square matrices only.
- An upper triangular matrix is a square matrix with non-zero entries only above or on the main diagonal (the left to right diagonal).
- A lower triangular matrix is a square matrix with non-zero entries only below or on the main diagonal.

Example 1.10

$$X = \begin{bmatrix} 5 & 2 & 7 \\ 8 & 1 & 3 \\ 2 & 7 & 4 \end{bmatrix}$$

While keeping the values intact, wherever possible, convert **X** into:

- A. a diagonal matrix
- B. an upper triangular matrix

C. a lower triangular matrix

Part A

A diagonal matrix has non zero values only on the diagonal. The remaining values must become zero:

$$\begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Part B

All values below the main diagonal must be converted to zero:

$$\begin{bmatrix} 5 & 2 & 7 \\ 0 & 1 & 3 \\ 0 & 0 & 4 \end{bmatrix}$$

Part C

All values above the main diagonal must be converted to zero:

$$\begin{bmatrix} 5 & 0 & 0 \\ 8 & 1 & 0 \\ 2 & 7 & 4 \end{bmatrix}$$

1.11: Identity Matrix

If in a diagonal matrix, all the non-zero elements are 1, then the matrix is called the identity matrix.

By convention, we use the variable I for an identity matrix.

$$I_1 = [1]$$

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

In general,

$$I_n = n \times n \text{ identity matrix}$$

The identity matrix serves the role of the multiplicative identity in the matrix system. We will see this property in action when we do multiplication of matrices later on.

Recall that in the real numbers, the multiplicative identity is given by:

$$x \times a = x \Rightarrow a = 1, \quad x \neq 0$$

1.12: Zero Matrix

If in a matrix, all the elements are 0, then the matrix is a zero matrix.

Zero matrices, unlike identity matrices, are not necessarily square matrices

A zero matrix takes the form

$$O_{m \times n}, \quad m \text{ rows}, n \text{ columns}$$

Note the use of the letter O , which reminds you of zero.

Example 1.13

Write the following

A. $O_{2 \times 2}$ B. $O_{3 \times 2}$

$$O_{2 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$O_{3 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

1.14: Row and Column Vector

- A matrix with one row and n columns is a n -dimensional row vector.
- A matrix with n rows and one column is a n -dimensional column vector.

Example 1.15

Find the order of the matrices below. Also, state whether they are row vectors, or column vectors.

A. $\begin{bmatrix} 3 \\ 4 \\ \frac{1}{2} \\ 2 \end{bmatrix}$

B. $\begin{bmatrix} \frac{1}{2} & \frac{1}{3} & -\frac{1}{4} & \frac{1}{5} \end{bmatrix}$

Order: 3×1 , Column Vector

Order: 1×4 , Row Vector

B. Transpose of a Matrix

1.16: Transpose of a Matrix

The transpose of a matrix is found by interchanging the rows and columns of the matrix.

$$\text{Transpose of } A = A^T$$

Example 1.17

Find the transpose of the matrices below:

A. $\begin{bmatrix} 3 \\ 4 \\ \frac{1}{2} \\ 2 \end{bmatrix}$

B. $\begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$

$$\begin{bmatrix} 3 & 4 & \frac{1}{2} \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 2 & 3 \\ 3 & 4 \\ 4 & 5 \end{bmatrix}$$

Example 1.18

- A. Find the number of rows and columns of A^T given that $A_{3 \times 7}$ is a matrix.

$$A_{7 \times 3}^T$$

1.19: Transpose “cancels”

$$(A^T)^T = A$$

- The transpose of a matrix is a self-inverse function.

C. Equality of Matrices**1.20: Equality of a Matrix**

Two matrices are equal *if and only if*:

- They have the same order
- The elements in the same position are equal.

Example 1.21

Find the value of x if:

$$\begin{bmatrix} a+2 & a+3 \\ z+2 & z \end{bmatrix} = \begin{bmatrix} b-3 & 7-b \\ b & x+7 \end{bmatrix}$$

$$a+2 = b-3 \Rightarrow a-b = -5$$

$$a+3 = 7-b \Rightarrow a+b = 4$$

$$2a = -1 \Rightarrow a = -\frac{1}{2} \Rightarrow b = 4 - a = 4 - \left(-\frac{1}{2}\right) = 4.5$$

$$z+2 = b \Rightarrow z = b-2 = 4.5-2 = 2.5$$

$$x+7 = z \Rightarrow x = z-7 = 2.5-7 = -4.5$$

Example 1.22

Find the product of the solutions for x if:

$$\begin{bmatrix} x & x \\ \sqrt{x} & x^x \end{bmatrix} = \begin{bmatrix} x^2 & |x| \\ x & x \end{bmatrix}$$

$$x = x^2$$

$$x = 0 \Rightarrow \text{Satisfies}$$

$$x \neq 0 \Rightarrow x = 1$$

$$x = |x| \Rightarrow x \geq 0$$

$$\sqrt{x} = x \Rightarrow x \in \{0,1\}$$

$$x^x = x \Rightarrow \mathbf{0^0 \text{ is not defined} \Rightarrow x = 1}$$

D. Number of Elements**Example 1.23**

What is the number of elements of:

- A. An $m \times n$ matrix

mn

1.2 Addition, Subtraction and Scalar Multiplication

A. Addition and Subtraction

1.24: Addition and Subtraction

Addition is defined only for two matrices that have the same order. That is, they have the same number of rows and the same number of columns.

$$a_{ij} \in A, b_{ij} \in B \Rightarrow A + B = a_{ij} + b_{ij}$$

Example 1.25

Given that $X = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $Y = \begin{bmatrix} 5 & 7 \\ 8 & 1 \end{bmatrix}$ find

- A. $X + Y$
- B. $X - Y$

$$X + Y = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 7 \\ 8 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 9 \\ 11 & 5 \end{bmatrix}$$

$$X - Y = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 7 \\ 8 & 1 \end{bmatrix} = \begin{bmatrix} -4 & -5 \\ -5 & 3 \end{bmatrix}$$

Example 1.26

Find A if the number of matrices in the first addition shown n terms:

$$\begin{bmatrix} 1 & 1 \\ \frac{1}{2} & 0 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ \frac{1}{4} & 0 \end{bmatrix} + \begin{bmatrix} 3 & 1 \\ \frac{1}{8} & 0 \end{bmatrix} + \dots = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = A$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \frac{n(n+1)}{2} & n \\ \frac{2^n - 1}{2^n} & 0 \end{bmatrix}$$

$$a = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$c = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n}$$

This is a geometric series with $a = \frac{1}{2}$, $r = \frac{1}{2}$ with sum:

$$S = \frac{a(1-r^n)}{1-r} = \frac{\frac{1}{2}\left(1-\left(\frac{1}{2}\right)^n\right)}{1-\frac{1}{2}} = \frac{\frac{1}{2}\left(1-\left(\frac{1}{2}\right)^n\right)}{\frac{1}{2}} = 1 - \frac{1}{2^n} = \frac{2^n - 1}{2^n}$$

B. Scalar Multiplication

In matrices, we will look at two kinds on multiplication. The first kind of multiplication multiplies a matrix with a scalar, while the second kind multiplies a matrix with a matrix. The second kind is more complicated.

1.27: Scalar Multiplication

Given a matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, and a scalar k , a scalar multiple of the matrix is given by:

$$kA = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

In other words, to multiply a matrix by a scalar k , we multiply each element of the matrix by k .

Example 1.28

John's Groceries bought the following at wholesale prices:

Rice at Rs. 95 per kg
Wheat at Rs. 85 per kg
Salt at Rs. 30 per kg

They sold at the following retail prices:

Rice at Rs. 100 per kg
Wheat at Rs. 87 per kg
Salt at Rs. 40 per kg

- A. Write a column matrix C with the cost of the items.
- B. Write a column matrix R with the selling price of the items
- C. Calculate $P = R - C$ and interpret it.
- D. John's Groceries bought and sold 30 kg of each item on Wed, 5th June. Calculate $30P$ and interpret it.

$$P = R - C = \begin{bmatrix} 100 \\ 87 \\ 40 \end{bmatrix} - \begin{bmatrix} 95 \\ 85 \\ 30 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 10 \end{bmatrix}$$

P represents the profit per kg of each item.

$$30P = 30 \begin{bmatrix} 5 \\ 2 \\ 10 \end{bmatrix} = \begin{bmatrix} 150 \\ 60 \\ 300 \end{bmatrix}$$

$30P$ represents the total profit for 30 kg of each item.

1.3 Matrix Multiplication

A. Matrix Multiplication

1.29: Multiplying Two Matrices

For matrix multiplication to be valid, the number of columns in the first matrix must equal the number of rows in the second matrix.

$$A = \underbrace{m}_{\text{Rows}} \times \underbrace{n}_{\text{Columns}}, \quad B = \underbrace{n}_{\text{Rows}} \times \underbrace{p}_{\text{Columns}}$$

1.30: Multiplying Two Matrices

To multiply two matrices, you need a:

$$A = \underbrace{m}_{\text{Rows}} \times \underbrace{n}_{\text{Columns}}, \quad B = \underbrace{n}_{\text{Rows}} \times \underbrace{p}_{\text{Columns}} \Rightarrow AB = \underbrace{m}_{\text{Rows}} \times \underbrace{p}_{\text{Columns}}$$

$$A_{m \times n} B_{n \times p} = (AB)_{m \times p}$$

- The number of columns in A is equal to the number of rows in the B .
- AB has the same number of rows as A and the same number of columns as B .

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$$

$$\mathbf{C} = \mathbf{AB} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix}$$

$\mathbf{C}_{11} = \mathbf{C} \begin{matrix} \underbrace{1}_{\text{Row Number}} & \underbrace{1}_{\text{Column Number}} \end{matrix} \Rightarrow$ 1st Row from A and 1st Column from B:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} p \\ r \end{bmatrix} = (a, b) \cdot \begin{pmatrix} p \\ r \end{pmatrix} = ap + br$$

$\mathbf{C}_{12} \Rightarrow$ 1st Row from A and 2nd Column from B:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} q \\ s \end{bmatrix} = aq + bs$$

$\mathbf{C}_{21} \Rightarrow$ 2nd Row from A and 1st Column from B:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} p \\ r \end{bmatrix} = cp + dr$$

$\mathbf{C}_{22} \Rightarrow$ 2nd Row from A and 2nd Column from B:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} q \\ s \end{bmatrix} = cq + ds$$

$$\mathbf{C} = \mathbf{AB} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{bmatrix}$$

Example 1.31

1. Find $\mathbf{AB}$ given that $\mathbf{A} = \begin{bmatrix} 3 & 6 \\ 0 & 2 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 1 & 4 \\ 2 & 5 \end{bmatrix}$
2. Find $\mathbf{AB}$ given that $\mathbf{A} = \begin{bmatrix} \frac{1}{2} & -2 \\ 0.3 & 0 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$
3. Find a, b, c, d given that $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$

Part 1

$$\mathbf{AB} = \begin{bmatrix} 3(1) + 6(2) & 3(4) + 6(5) \\ 0(1) + 2(2) & 0(4) + 2(5) \end{bmatrix} = \begin{bmatrix} 3 + 12 & 12 + 30 \\ 0 + 4 & 0 + 10 \end{bmatrix} = \begin{bmatrix} 15 & 42 \\ 4 & 10 \end{bmatrix}$$

Part 2

$$\mathbf{AB} = \begin{bmatrix} \frac{1}{2} & -4 \\ -\frac{1}{2} & -4 \\ 0.9 & 0 \end{bmatrix}$$

Part 3

$$\begin{bmatrix} a + 3b & 2a + 4b \\ c + 3d & 2c + 4d \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$$

$$a + 3b = 4 \Rightarrow \underbrace{2a + 6b = 8}_{\text{Equation I}}$$

$$\underbrace{2a + 4b = 3}_{\text{Equation II}}$$

$$2b = 5 \Rightarrow b = \frac{5}{2}$$

Substitute $b = \frac{5}{2}$ in $a + 3b = 4$:

$$a + 3\left(\frac{5}{2}\right) = 4 \Rightarrow a = 4 - \frac{15}{2} = -\frac{7}{2}$$

Example 1.32

If $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$, and $A^2 = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$, then determine the value of α and β in terms of a and b : (JEE-M 2003)

Apply the definition:

$$A^2 = AA = \begin{bmatrix} a & b \\ b & a \end{bmatrix} \begin{bmatrix} a & b \\ b & a \end{bmatrix} = \begin{bmatrix} a^2 + b^2 & ab + ba \\ ba + ab & b^2 + a^2 \end{bmatrix} = \begin{bmatrix} a^2 + b^2 & 2ab \\ 2ab & a^2 + b^2 \end{bmatrix} = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$$

Comparing gives us:

$$\alpha = a^2 + b^2, \beta = 2ab$$

Example 1.33

Mark the correct option

If $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ then which one of the following holds for all $n \geq 1$ by the principle of mathematical induction: (JEE-M 2005)

- A. $A^n = nA - (n-1)I$
- B. $A^n = 2^{n-1}A - (n-1)I$
- C. $A^n = nA + (n-1)I$
- D. $A^n = 2^{n-1}A + (n-1)I$

$$A^2 = AA = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1+0 & 0+0 \\ 1+1 & 0+1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

$$A^3 = A^2A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1+0 & 0 \\ 2+1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$$

We can prove using mathematical induction. Assume that $A^n = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$ and show that:

$$A^{n+1} = A^n A = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1+0 & 0 \\ n+1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ n+1 & 1 \end{bmatrix}$$

$$A^n = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$$

Using $n = n - (n - 1)$:

$$= \begin{bmatrix} n - (n-1) & 0 \\ n & n - (n-1) \end{bmatrix} = \begin{bmatrix} n & 0 \\ n & n \end{bmatrix} - \begin{bmatrix} n-1 & 0 \\ 0 & n-1 \end{bmatrix}$$

Factor n out of the first matrix, and $n-1$ out of the second matrix and:

$$= n \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} - (n-1) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{Substitute } A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$= nA - (n-1)I \Rightarrow \text{Option A}$$

Example 1.34

If A and B are square matrices of size $n \times n$ such that $A^2 - B^2 = (A - B)(A + B)$ then which of the following will always be true? (JEE-M 2006)

- A. $A = B$

- B. $AB = BA$
- C. Either of A or B is a zero matrix
- D. Either of A or B is an identity matrix

$$\begin{aligned}
 A^2 - B^2 &= (A - B)(A + B) \\
 A^2 - B^2 &= A^2 + AB - BA - B^2 \\
 0 &= AB - BA \\
 AB &= BA \\
 \text{Option B}
 \end{aligned}$$

Example 1.35

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ such that $AB = BA$, then, the number of matrices B that meet the given condition is: (JEE-M 2006)

$$\begin{aligned}
 AB &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} = \begin{bmatrix} a+0 & 2b \\ 3a+0 & 0+4b \end{bmatrix} = \begin{bmatrix} a & 2b \\ 3a & 4b \end{bmatrix} \\
 BA &= \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} a+0 & 2a+0 \\ 0+3b & 0+4b \end{bmatrix} = \begin{bmatrix} a & 2a \\ 3b & 4b \end{bmatrix}
 \end{aligned}$$

Comparing elements:

$$\begin{aligned}
 2a = 2b, \Leftrightarrow 3a = 3b \Leftrightarrow 4a = 4b \Leftrightarrow a = b \\
 \text{Infinite solutions} \Rightarrow \text{Infinite matrices}
 \end{aligned}$$

Example 1.36: Complex Numbers

If $\omega \neq 1$ is the complex cube root of unity and matrix $H = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$, then H^{70} in terms of H is: (JEE-M 2011)

We will make use of the property:

$$\omega = \sqrt[3]{1} \Rightarrow \omega^3 = 1$$

$$\begin{aligned}
 H^2 &= \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix} \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix} = \begin{bmatrix} \omega^2 + 0 & 0 + 0 \\ 0 + 0 & 0 + \omega^2 \end{bmatrix} = \begin{bmatrix} \omega^2 & 0 \\ 0 & \omega^2 \end{bmatrix} \\
 H^n &= \begin{bmatrix} \omega^n & 0 \\ 0 & \omega^n \end{bmatrix}
 \end{aligned}$$

$$H^{70} = \begin{bmatrix} \omega^{70} & 0 \\ 0 & \omega^{70} \end{bmatrix}$$

Substitute $\omega^{70} = \omega^{69} \cdot \omega = (\omega^3)^{23} \cdot \omega = (1)^{23} \cdot \omega = 1 \cdot \omega = \omega$:

$$H^{70} = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix} = H$$

Hence,

$$H^{70} = H$$

Example 1.37

- A. My weekly grocery shopping list comprises 3 packs of apple juice, 4 packets of breakfast cereal, and 2 packets of my favorite biscuit. Write these quantities as row matrix Q .
- B. Starbazaar offers apple juice at Rs. 95 per pack, breakfast cereal at Rs. 30 per pack, and a biscuit packet

at Rs. 12 per packet. DMart offers apple juice at Rs. 100 per pack, breakfast cereal at Rs. 27 per pack, and a biscuit packet at Rs. 10 per packet. Write these quantities as column matrix $\mathbf{C}$ with two columns

- C. Determine $\mathbf{QC}$ using matrix multiplication.
- D. Interpret $\mathbf{QC}$.
- E. I can purchase the items from one place. From where should I purchase and why?

Part A

$$\text{Quantity Matrix} = \mathbf{Q} = [3 \quad 4 \quad 2]_{1 \times 3}$$

Part B

$$\text{Cost Matrix} = \mathbf{C} = \begin{bmatrix} 95 & 100 \\ 30 & 27 \\ 12 & 10 \end{bmatrix}_{3 \times 2}$$

Part C

First, check the feasibility:

$$Q_{1 \times 3}, C_{3 \times 2}$$

Number of Columns of $\mathbf{Q}$ is same as number of rows of $\mathbf{C}$.

The answer matrix will have

- Same number of Rows as $\mathbf{Q} = 1$
- Same number of Columns as $\mathbf{C} = 2$

$$\mathbf{QC} = [3 \quad 4 \quad 2]_{1 \times 3} \begin{bmatrix} 95 & 100 \\ 30 & 27 \\ 12 & 10 \end{bmatrix}_{3 \times 2} = [a_{11} \quad a_{12}]$$

$$= [3 \times 95 + 4 \times 30 + 2 \times 12 \quad 3 \times 100 + 4 \times 27 + 2 \times 10] = [429 \quad 428]$$

Part D

$$\mathbf{QC} = [429 \quad 428]$$

The first entry represents the cost if items are purchased at Starbazaar.

The second entry represents the cost if items are purchased at DMart.

Part E

You should purchase from Dmart, because you will save:

$$429 - 428 = 1 \text{ Rs.}$$

B. Identifying the Order

Example 1.38

Each part below gives two matrices $\mathbf{A}$ and $\mathbf{B}$. Decide whether $\mathbf{AB}$ is possible. Also, decide $\mathbf{BA}$ is possible. For matrices which can be multiplied, give the order of the product of the two matrices.

1. $\mathbf{A}$ is a 3×4 matrix. $\mathbf{B}$ is a 4×2 matrix.
2. $\mathbf{A}$ is a 4×4 matrix. $\mathbf{B}$ is a 4×6 matrix.
3. $\mathbf{A}$ is a $x \times y$ matrix. $\mathbf{B}$ is a $y \times x$ matrix.
4. $\mathbf{A}$ is a 2×3 matrix. $\mathbf{B}$ is a 3×2 matrix.
5. $\mathbf{A}$ is a 5×9 matrix. $\mathbf{B}$ is a 5×11 matrix.

Part 1

$\mathbf{AB}$ has order 3×2

$\mathbf{BA}$ is not possible

Part 2

AB has order 4×6

BA is not possible

Part 3

AB has order $x \times x$

BA has order $y \times y$

Part 4

AB has order 2×2

BA has order 3×3

Part 5

AB is not possible

BA is not possible

C. Commutative Property

1.39: Commutative Property in Multiplication for Real Numbers

The commutative property for real numbers says that:

$$a \times b = b \times a$$

1.40: Commutative Property in Matrix Multiplication

Matrix multiplication, in general, is not commutative.

$$\mathbf{AB} \neq \mathbf{BA}$$

- It is not necessary if $\mathbf{AB}$ exists, that $\mathbf{BA}$ exists
- Even if it does exist, it need not be the same.

Example 1.41

Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}, \mathbf{X} = \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix}$$

- A. Determine $\mathbf{AX}$.
- B. Determine $\mathbf{XA}$.
- C. Are the two equal?

$$\begin{aligned} \mathbf{AX} &= \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 2+0 & 1+0 \\ 4+0 & 2+0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} \\ \mathbf{XA} &= \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 2+2 & 0+0 \\ 3+0 & 0+0 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 3 & 0 \end{bmatrix} \end{aligned}$$

Example 1.42

Consider two matrices $\mathbf{X}$ and $\mathbf{Y}$, where $\mathbf{XY}$ is defined. State always true or always false, or sometimes true and sometimes false.

- A. $\mathbf{YX}$ exists.
- B. If $\mathbf{YX}$ exists, then $\mathbf{YX} = \mathbf{XY}$

Part A

$\mathbf{A}$ is a 3×4 matrix. $\mathbf{B}$ is a 4×2 matrix.

$\mathbf{A}$ is a $x \times y$ matrix. $\mathbf{B}$ is a $y \times x$ matrix.

Sometimes true, sometimes false.

Part B

Suppose

$$Y = X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow XY = YX = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \text{Sometimes true}$$

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}, X = \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix} \Rightarrow AX \neq XA \Rightarrow \text{Sometimes false}$$

Sometimes true, sometimes false.

D. Multiplicative Identity**1.43: Multiplicative Identity for Matrices**

$$IA = AI = A$$

The identity matrix (covered earlier) serves the role of 1 in matrices. Multiplying any matrix by I does not change it.

$$AI = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a \times 1 + b \times 0 & a \times 0 + b \times 1 \\ c \times 1 + d \times 0 & c \times 0 + d \times 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$IA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 \times a + 0 \times c & 1 \times b + 0 \times d \\ 0 \times a + 1 \times c & 0 \times b + 1 \times d \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Example 1.44

Show that

$$AI_2 = A$$

We need to show this for any matrix A , so we take the most general value of A , without any restrictions.
Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Then,

$$LHS = AI_2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a \times 1 + b \times 0 & a \times 0 + b \times 1 \\ c \times 1 + d \times 0 & c \times 0 + d \times 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = A = RHS$$

1.45: Powers of the Multiplicative Identity

$$I^n = I, n \in \mathbb{N}$$

$$I^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$I^n = \underbrace{IIII \dots II}_{n \text{ times}}$$

Since $I^2 = I$, we can simplify the last two matrices as I :

$$I^n = \underbrace{IIII \dots II}_{n-1 \text{ times}}$$

And if we keep repeating the above process, we will eventually get:

$$I^n = I, n \in \mathbb{N}$$

1.46: Multiplicative Identity for Matrices (Proof by Induction)

$$I^n = I, n \in \mathbb{N}$$

For $n = 1$:

$$I^1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Base Case

Prove the base case.

For $n = 2$:

$$I^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Inductive Step

Let

$$I^n = I \text{ for some } n$$

Then

$$I^{n+1} = II^n = II = I^2 = I$$

Hence, by induction:

$$I^n = I, n \in \mathbb{N}$$

E. Zero Matrix**1.47: Zero Matrix**

$$OA = AO = O$$

$$O_2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Anything multiplied by the zero matrix results in a zero.

$$AO = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a \times 0 + b \times 0 & a \times 0 + b \times 0 \\ c \times 0 + d \times 0 & c \times 0 + d \times 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$OA = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 0 \times a + 0 \times c & 0 \times b + 0 \times d \\ 0 \times a + 0 \times c & 0 \times b + 0 \times d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

1.48: Distributive Property

$$A(B + C) = AB + AC$$

Let:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} p & q \\ r & s \end{bmatrix}, \quad C = \begin{bmatrix} w & x \\ y & z \end{bmatrix}$$

Then:

$$LHS = A(B + C) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} p+w & q+x \\ r+y & s+z \end{bmatrix} = \begin{bmatrix} a(p+w) + b(r+y) & a(q+x) + b(s+z) \\ c(p+w) + d(r+y) & c(q+x) + d(s+z) \end{bmatrix}$$

$$AB = \begin{bmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{bmatrix}$$

$$AC = \begin{bmatrix} aw + by & ax + bz \\ cw + dy & cx + dz \end{bmatrix}$$

Then the **RHS**:

$$AB + AC = \begin{bmatrix} ap + br + aw + by & aq + bs + ax + bz \\ cp + dr + cw + dy & cq + ds + cx + dz \end{bmatrix} = \begin{bmatrix} a(p+w) + b(r+y) & a(q+x) + b(s+z) \\ c(p+w) + d(r+y) & c(q+x) + d(s+z) \end{bmatrix} = LHS$$

Hence, proved.

1.49: Associative Property of Multiplication

$$(AB)C = A(BC)$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} p & q \\ r & s \end{bmatrix}, \quad C = \begin{bmatrix} w & x \\ y & z \end{bmatrix}$$

$$(AB)C = \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix} \right) \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{bmatrix} \begin{bmatrix} w & x \\ y & z \end{bmatrix} =$$

$$A(BC) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \left(\begin{bmatrix} p & q \\ r & s \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix} \right) =$$

1.50: Powers of a Matrix

If the matrix multiplication is defined, then:

$$\begin{aligned} A^2 &= AA \\ A^3 &= AAA \\ A^n &= \underbrace{AA \dots A}_{n \text{ times}} \end{aligned}$$

For the matrix multiplication to be defined,

No. of Rows

F. Matrix Equations

Because the commutative property does not hold for matrices, the order of multiplication in matrices matters.

1.51: Pre-Multiplication and Post Multiplication

Consider matrix A .

- If you multiply with a matrix before A , then it is called pre-multiplication.
- If you multiply with a matrix after A , then it is called post-multiplication.

Example 1.52

$$A = B$$

- A. Premultiply both sides by X .
- B. Postmultiply both sides by X .

$$XA = XB$$

$$AX = BX$$

Example 1.53

If $M = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$, then find the value of M^n .

Note that

$$M = -\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = -I \Rightarrow M^n = (-I)^n = (-1)^n I$$

Which is:

$$\begin{aligned} &I, \text{ when } n \text{ is even} \\ &-I, \text{ when } n \text{ is odd} \end{aligned}$$

1.4 Inverse Matrices

A. Multiplicative Inverse

1.54: Multiplicative Inverse for Real Numbers

If a and b are real numbers and

$$ab = 1 \Rightarrow a = \frac{1}{b} \Rightarrow b = \frac{1}{a}$$

Then:

- b is called the multiplicative inverse of a
- a is called the multiplicative inverse of b

$$\frac{2}{3} = 2 \div 3 = 2 \times \frac{1}{3}$$

$\frac{1}{3}$ is the multiplicative inverse of 3 because:

$$3 \times \frac{1}{3} = 1$$

1.55: Multiplicative Inverse for Matrices

Given a matrix A , the multiplicative inverse of the matrix is A^{-1} is a matrix that satisfies the conditions:

$$AA^{-1} = A^{-1}A = I$$

Where:

$$I = \text{Identity Matrix}$$

In order for a matrix to have an inverse:

- It must be a square matrix

Example 1.56

Show that the matrix $A = \begin{bmatrix} -2 & 1 \\ 3 & -\frac{1}{2} \end{bmatrix}$ is the inverse of the matrix $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ using matrix multiplication.

We can show that:

$$BA = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 3 & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} -2+3 & 1-1 \\ -6+6 & 3-2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

$$AB = \begin{bmatrix} -2 & 1 \\ 3 & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} -2+3 & -4+4 \\ \frac{3}{2}-\frac{3}{2} & 3-2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

1.57: Finding the Inverse

Given a matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the inverse of A , denoted A^{-1} is given by:

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example 1.58

Find the inverse of each matrix below, if it exists.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 11 \\ 1 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 6 & 8 \\ 3 & 4 \end{bmatrix}, \quad D = \begin{bmatrix} 7 & 3 \\ 9 & 4 \end{bmatrix}, \quad E = \begin{bmatrix} -1 & 2 \\ 4 & 7 \end{bmatrix}$$

$$A^{-1} = \frac{1}{4-6} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 3 & -1 \\ 2 & -2 \end{bmatrix}$$

$$B^{-1} = \frac{1}{12-11} \begin{bmatrix} 4 & -11 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 4 & -11 \\ -1 & 3 \end{bmatrix}$$

$$C^{-1} = \frac{1}{24-24} \begin{bmatrix} 4 & -8 \\ -3 & 6 \end{bmatrix} \Rightarrow \text{Inverse does not exist}$$

$$D^{-1} = \frac{1}{28-27} \begin{bmatrix} 4 & -3 \\ -9 & 7 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -9 & 7 \end{bmatrix}$$

$$E^{-1} = \frac{1}{(-1)(7) - (2)(4)} \begin{bmatrix} 7 & -2 \\ -4 & -1 \end{bmatrix} = \frac{1}{-15} \begin{bmatrix} 7 & -2 \\ -4 & -1 \end{bmatrix} = \begin{bmatrix} -\frac{7}{15} & \frac{2}{15} \\ \frac{4}{15} & \frac{1}{15} \end{bmatrix}$$

B. Existence of Inverses

1.59: Singular Matrices

If the quantity $ad - bc$ for a 2×2 matrix is zero, then the matrix does not have an inverse. Such matrices are called singular.

The converse is also true. If a matrix is singular, then the quantity $ad - bc = 0$

(Calculator Allowed) Example 1.60

Find the exact values of x for which the matrix given below is singular:

$$\begin{bmatrix} 3x+4 & 5x+9 \\ 4x+3 & 2x-7 \end{bmatrix}$$

$$\begin{aligned} (3x+4)(2x-7) &= (4x+3)(5x+9) \\ 6x^2 - 21x + 8x - 28 &= 20x^2 + 36x + 15x + 27 \\ 0 &= 14x^2 + 64x + 55 \end{aligned}$$

$$x = \frac{-64 \pm \sqrt{4096 - (4)(14)(55)}}{28} = \frac{-32 \pm \sqrt{254}}{14}$$

1.61: Non-Singular Matrices

If the quantity $ad - bc$ for a 2×2 matrix is non-zero, then the matrix has an inverse. Such matrices are called invertible, or non-singular.

Example 1.62

If $A = \begin{bmatrix} a & 7 \\ 4 & 3 \end{bmatrix}$ is a singular matrix and $a \in \mathbb{R}$, then determine the value of a .

$$3a - 7 \times 4 = 0 \Rightarrow 3a - 28 = 0 \Rightarrow 3a = 28 \Rightarrow a = \frac{28}{3}$$

Example 1.63

If $A = \begin{bmatrix} a & v \\ v & a \end{bmatrix}$ is a singular matrix and $a, v \in \mathbb{R}$, then is it true that $a = v$?

$$a^2 - v^2 = 0 \Rightarrow a^2 = v^2 \Rightarrow a = \pm v$$

Hence, the statement above is true, but not complete.

Example 1.64

Find the inverse of the following matrix, and state the values of k for which A^{-1} exists

$$\begin{bmatrix} k^2 & k-1 \\ 2k & 1 \end{bmatrix}$$

$$\frac{1}{(k^2)(1) - (2k)(k-1)} \begin{bmatrix} 1 & 1-k \\ -2k & k^2 \end{bmatrix}$$

The denominator of the fraction cannot be zero. Hence, we find the zeroes of the denominator:

$$(k^2)(1) - (2k)(k-1) = 0$$

$$k^2 - 2k^2 + 2k = 0$$

$$-k^2 + 2k = 0$$

$$k(-k + 2) = 0$$

$$k \in \{0, 2\}$$

The inverse matrix exists for

$$k \in \mathbb{R} - \{0, 2\}$$

Example 1.65

$$(kA) \left(\frac{1}{k} A^{-1} \right)$$

$$(kA) \left(\frac{1}{k} A^{-1} \right) = \left(k \times \frac{1}{k} \right) (A \times A^{-1}) = (1)(I) = I$$

C. Inverse of an Inverse

1.66: Inverse of an Inverse

The inverse of an inverse matrix is the original matrix.

$$(X^{-1})^{-1} = X$$

Method I

$$A^{-1}A = I$$

Take the inverse both sides. Note that I^{-1} is just I . And $(A^{-1}A)^{-1} = (A^{-1})^{-1}A^{-1}$

$$(A^{-1})^{-1}A^{-1} = I$$

$$(A^{-1})^{-1}A^{-1}A = A$$

$$\begin{aligned}(A^{-1})^{-1}I &= A \\ (A^{-1})^{-1} &= A\end{aligned}$$

Method II

Consider the expression $(X^{-1})^{-1}(X^{-1})$

Use a change of variable. Let $X^{-1} = Y$:

$$(X^{-1})^{-1}(X^{-1}) = (Y)^{-1}(Y) = I$$

But, the only matrix that will multiply with X^{-1} to give I is X . Hence,

$$(X^{-1})^{-1} = X$$

The same will work with

$$(X^{-1})(X^{-1})^{-1}$$

D. Self-Inverse Matrices

1.67: Self Inverse Matrices

A self-inverse matrix is a matrix such that it is its own inverse.

$$A = A^{-1}$$

1.68: Self Inverse Matrices

A 2×2 matrix is self-inverse *if and only if*:

$$A = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}, \quad a^2 + bc = 1$$

Where a, b, c need not be distinct.

Let

$$A = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$$

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} \frac{d}{ad - bc} & -\frac{b}{ad - bc} \\ -\frac{c}{ad - bc} & \frac{a}{ad - bc} \end{bmatrix}$$

Since the matrix is self-inverse, it must be equal to its own inverse. Equate elements of the matrices:

$$a_{12}: b = -\frac{b}{ad - bc} \Rightarrow ad - bc = -1$$

$$a_{21}: c = -\frac{c}{ad - bc} \Rightarrow ad - bc = -1$$

$$a_{11}: a = \frac{d}{ad - bc} = \frac{d}{-1} = -d$$

$$a_{22}: d = \frac{a}{ad - bc} = \frac{a}{-1} = -a$$

Substituting $a = -d$ into $ad - bc = -1$ gives:

$$ad - bc = -a^2 - bc = -1 \Rightarrow a^2 + bc = 1 \Rightarrow A = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$$

And we can confirm that its inverse is:

$$A^{-1} = \frac{1}{-a^2 - bc} \begin{bmatrix} -a & -b \\ -c & a \end{bmatrix} = \frac{1}{-1} \begin{bmatrix} -a & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} a & b \\ c & -a \end{bmatrix} = A$$

1.69: Square of a Self-Inverse Matrix

If a matrix A is a self-inverse matrix, then

$$A^2 = I$$

Since A is a self-inverse matrix, we must have:

$$A = A^{-1}$$

Post-multiply both sides by A :

$$\begin{aligned} AA &= A^{-1}A \\ A^2 &= I \end{aligned}$$

E. Idempotent Matrix

Example 1.70

I have a real number x . Determine the domain of $f(x) = \frac{1}{x}$.

$$x \in \mathbb{R}, \quad x \neq 0$$

Example 1.71

Consider matrices A and B such that $AB = B$, $BA = A$. Explain what is wrong with the steps below which conclude that both A and B are identity matrices.

Pre-multiply by A^{-1} :

$$A^{-1}AB = A^{-1}B \Rightarrow IB = A^{-1}B \Rightarrow B = A^{-1}B$$

Post Multiply by B^{-1} on both sides:

$$BB^{-1} = A^{-1}BB^{-1} \Rightarrow I = A^{-1}I \Rightarrow I = A^{-1}$$

Take the inverse both sides:

$$I^{-1} = A \Rightarrow I = A$$

Using a similar set of steps, we can conclude that:

$$I = B$$

And hence:

$$I = A = B$$

The information does not specify that the matrices A and B are invertible. Hence, the operations are only valid, if the matrices are invertible.

Example 1.72

Given that $AB = B$ and $BA = A$ show that $A^2 = A$.

Since the matrices are not known to be invertible, we cannot, in general, multiply by inverse matrices:

$$AB = B$$

Post Multiply both sides by A

$$ABA = BA$$

Substitute $BA = A$

$$AA = A$$

$$A^2 = A$$

1.73: Idempotent Matrix

A matrix such it is equal to its square is an idempotent matrix. For an idempotent matrix A , we have the condition:

$$A^2 = A$$

Example 1.74

Explain what is wrong with the steps below, which show that an idempotent matrix is either a zero matrix or an identity matrix.

$$A^2 = A$$

$$A^2 - A = 0$$

$$A(A - I) = 0$$

$$A = 0 \text{ or } A = I$$

$$A^2 - A = 0$$

$$A(A^2 A^{-1} - A^{-1}) = 0$$

At the stage above, we are multiplying by the inverse, which may or may not exist. Hence, that step is wrong.

$$A(A - I) = 0$$

Hence, the step is only correct if A has an inverse.

$$3x + 6y = 3\left(\frac{3x}{x} + \frac{6y}{3}\right)$$

1.75: Idempotent Matrix

The condition for a 2×2 matrix to be idempotent is that:

Case I: If at least one of b and c is non-zero:

$$\underbrace{d = 1 - a}_{\text{Condition I}}, \quad \underbrace{\text{Determinant} = ad - bc = 0}_{\text{Condition II}}$$

Case II: If b and c are both zero:

$$d \in \{0,1\}, \quad a \in \{0,1\}$$

$$A^2 = A$$

$$LHS = A^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{bmatrix}$$

$$RHS = A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Equate elements

$$ab + bd = b$$

For $b \neq 0$:

$$a + d = 1 \Rightarrow \underbrace{d = 1 - a}_{\text{Condition I}}$$

For $c \neq 0$:

$$\begin{aligned} ac + cd &= c \\ a + d = 1 &\Rightarrow \underbrace{d = 1 - a}_{\text{Condition I}} \end{aligned}$$

$$\begin{aligned} a^2 + bc &= a \\ a^2 - a &= -bc \\ a(a - 1) &= -bc \\ a(1 - a) &= bc \\ ad &= bc \\ ad - bc &= 0 \\ \underbrace{\text{Determinant} = ad - bc}_{\text{Condition II}} &= 0 \end{aligned}$$

Suppose we relax the condition for b and c :

$b = 0, c \neq 0$:

$$\begin{aligned} ab + bd &= b \Rightarrow a(0) + b(0) = 0 \Rightarrow 0 = 0 \Rightarrow \text{Always True} \\ ac + cd &= c \Rightarrow a + d = 1 \Rightarrow \underbrace{d = 1 - a}_{\text{Condition I}} \end{aligned}$$

$b \neq 0, c = 0$:

$$\begin{aligned} ab + bd &= b \Rightarrow a + d = 1 \Rightarrow \underbrace{d = 1 - a}_{\text{Condition I}} \\ ac + cd &= c \Rightarrow a(0) + (0)d = 0 \Rightarrow \text{Always True} \end{aligned}$$

Hence, when one of b and c is non-zero, Condition I still applies.

$b = 0, c = 0$:

Substitute $b = c = 0$:

$$\begin{aligned} \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{bmatrix} &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \\ \begin{bmatrix} a^2 & 0 \\ 0 & d^2 \end{bmatrix} &= \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \\ a^2 &= a \Rightarrow a \in \{0, 1\} \\ d^2 &= d \Rightarrow d \in \{0, 1\} \end{aligned}$$

Example 1.76

$$a = 1, b = c = d = 0 \Rightarrow A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$A_1^2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \text{Idempotent}$$

$$d = 1, a = b = c = 0 \Rightarrow A_1 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A_1^2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \text{Idempotent}$$

Example 1.77

$$\begin{bmatrix} 3 & -6 \\ 1 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -6 \\ 1 & -2 \end{bmatrix}^2 = \begin{bmatrix} 3 & -6 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 3 & -6 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 9-6 & -18+12 \\ 3-2 & -6+4 \end{bmatrix} = \begin{bmatrix} 3 & -6 \\ 1 & -2 \end{bmatrix}$$

Check Condition I: $d = 1 - a$

$$d = 1 - a = 1 - 3 = -2 \Rightarrow \text{Valid}$$

Check Condition II: $ad - bc = 0$

$$3(-2) - (-6)(1) = -6 + 6 = 0 \Rightarrow \text{Valid}$$

Both the conditions above are met:

Matrix is idempotent

F. Nilpotent Matrix**1.78: Nilpotent Matrix**

A matrix is called a nilpotent matrix if some positive integer power of the matrix is the zero matrix. That is:

$$A^k = \mathbf{0}, \quad k \in \mathbb{N}$$

1.79: Nilpotent Matrix, $A^2 = \mathbf{0}$

$$\begin{aligned} A^2 &= \mathbf{0} \\ LHS = A^2 &= \begin{bmatrix} a & b \\ c & d \end{bmatrix}^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{bmatrix} \\ RHS = \mathbf{0} &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

$$ab + bd = 0 \Rightarrow b(a + d) = 0 \Rightarrow b = 0 \text{ OR } a = -d$$

$$ac + cd = 0 \Rightarrow c(a + d) = 0 \Rightarrow c = 0 \text{ OR } a = -d$$

$$a = -d \text{ OR } b = c = 0$$

If $b = c = 0$

$$a^2 + bc = 0 \Rightarrow a^2 = 0 \Rightarrow a = 0$$

$$bc + d^2 = 0 \Rightarrow d^2 = 0 \Rightarrow d = 0$$

Equate first and fourth equation:

$$a^2 + bc = bc + d^2$$

$$a^2 = d^2$$

$$a = \pm d$$

Example 1.80

$$\begin{bmatrix} a & b \\ c & -a \end{bmatrix}$$

1.5 Counting

A. Counting

1 Pending

Example 1.81

Find the number of matrices $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ such that A is a singular matrix and a, b, c, d are distinct natural numbers less than 10.

$$ad - bc = 0 \Rightarrow ad = bc$$

Example 1.82

How many 3×3 matrices have exactly four elements = 1 and remaining elements = 0.

We need to choose the positions of the 1's among the 9 elements of the 3×3 matrix:

$$\binom{9}{4}$$

2 Pending

Example 1.83

How many 3×3 non-singular matrices have exactly four elements = 1 and remaining elements = 0.

3 Pending

1.84: Trace

The trace of a square matrix is the sum of its diagonal elements.

Trace is not defined for non-square matrices.

B. Nilpotent Matrices

4 Pending

1.85: Trace and Determinant of a Nilpotent Matrix

For nilpotent matrices:

$$\begin{aligned}\text{Trace} &= 0 \\ \text{Determinant} &= 0\end{aligned}$$

5 Pending

Example 1.86

Does a nilpotent matrix have an inverse?

No. Because its determinant is zero.

6 Pending

Example 1.87

Determine number of nilpotent matrices with integer elements

Infinite since

$$x^2 + ab = 0$$

7 Pending

1.88: Binary Matrix

A binary matrix is a matrix that has only 0 and 1 for its elements.

8 Pending

Example 1.89

Determine number of 2 by 2 binary nilpotent matrices.

9 Pending

Example 1.90

Determine number of 2 by 2 nilpotent matrices with entries from $\{-1, 0, 1\}$

C. Permutation Matrix

10 Pending

1.91: Orthonormal Vectors

Orthonormal vectors are vectors which are perpendicular to each other.

11 Pending

1.92: Perpendicularity of Vectors: Dot Product Property

Two vectors are perpendicular if their dot product is zero.

12 Pending

1.93: Orthogonal Matrix

A square matrix is orthogonal if its columns and rows are orthonormal vectors.

13 Pending

1.94: Orthogonal Matrix

A matrix is orthogonal if the product of its transpose and itself, and the product of the matrix with the transpose, both equal the identity matrix.

$$Q(Q^T) = (Q^T)Q = I$$

14 Pending

1.95: Inverse of an Orthogonal Matrix

The inverse of an orthogonal matrix is also its transpose.

From the definition of inverse of matrix.

(Definition: A matrix B is an inverse of A if $AB = BA = I$)

15 Pending

1.96: Permutation Matrix

A square binary matrix with 1 entry of 1 in each column and each row and remaining entries zero.

16 Pending

1.97: Permutation Matrices are orthogonal

Permutation matrices are orthogonal. Hence:

- Permutation matrices are invertible.
- Permutation matrices have their transpose equal to their inverse.

D. Binary Matrices as Relations

17 Pending

1.98: Matrices as Relations

A binary matrix represents a relation.

18 Pending

Example 1.99

Count Number of $n \times n$ Binary Matrices. Interpret this matrix representation as a Relation

A. matrices are symmetric

$$2^{n^2}$$

Relation from set X with n elements to itself.

19 Pending

Example 1.100

Count Number of $n \times n$ Binary Matrices with trace n . Interpret this as a Relation

$$2^{n(n-1)}$$

This counts the number of reflexive relations from a set X with n elements to itself.

20 Pending

Example 1.101

Count Number of $n \times n$ Binary Symmetric Matrices. Interpret this as a Relation

$$2^{\frac{n(n+1)}{2}}$$

Symmetric Relation

1.6 Determinants

A. Determinants (2×2)

1.102: Determinant

The determinant of the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is

$$\det A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Notation:

- A matrix has brackets (often square, but can also round).
- A determinant has straight lines.

Example 1.103A. Evaluate $\begin{vmatrix} 2 & 5 \\ 1 & 3 \end{vmatrix}$ B. Find the value of b if the determinant of the matrix $\begin{bmatrix} b & 4 \\ 7 & 3 \end{bmatrix}$ is -2 .**Part A**

$$\begin{vmatrix} 2 & 5 \\ 1 & 3 \end{vmatrix} = (2)(3) - (5)(1) = 1$$

Part B

$$3b - 28 = -2 \Rightarrow 3b = 26 \Rightarrow b = \frac{26}{3}$$

Example 1.104If $\begin{vmatrix} 3 & 2c \\ c & 8 \end{vmatrix}$ is a positive integer, then find the sum of the possible values that it can take.

$$24 - 2c^2 = K$$

$$c = 0 \Rightarrow K = 24 - 0 = 24$$

$$c = \pm 1 \Rightarrow K = 24 - 2 = 22$$

$$c = \pm 2 \Rightarrow K = 24 - 8 = 16$$

$$c = \pm 3 \Rightarrow K = 24 - 18 = 6$$

$$c = \pm 4 \Rightarrow K = 24 - 32 = -8 < 0 \Rightarrow \text{Not Valid}$$

$$|c| \geq 4 \Rightarrow \text{Not Valid}$$

The sum of the possible values is:

$$24 + 22 + 16 + 6 = 68$$

Example 1.105If θ is an angle such it can be expressed in degrees $\theta = n, 0 \leq n \leq 360, n \in \mathbb{Z}$, find the sum of the values that satisfy

$$\begin{vmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{vmatrix} = 1$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\theta \in \{0, 1, 2, \dots, 360\}$$

$$\text{Sum} = 1 + 2 + \dots + 360 = \frac{360 \times 361}{2}$$

1.106: "Distributing" the Determinant

$$\det(\mathbf{AB}) = \det \mathbf{A} \times \det \mathbf{B}$$

Let $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \mathbf{B} = \begin{bmatrix} w & x \\ y & z \end{bmatrix}$. First,

$$RHS = \det \mathbf{A} \times \det \mathbf{B} = (ad - bc)(wz - xy) = adwz - adxy - bcwz + bcxy$$

$$\begin{aligned}
 LHS &= \det \mathbf{AB} = \begin{vmatrix} aw + by & ax + bz \\ cw + dy & cx + dz \end{vmatrix} \\
 &= (aw + by)(cx + dz) - (cw + dy)(ax + bz) \\
 &= \textcolor{violet}{acwx} + adwz + bcxy + \textcolor{violet}{bdyz} - \textcolor{violet}{acwx} - bcwz - adxy - \textcolor{violet}{bdyz} \\
 &= adwz + bcxy - bcwz - adxy
 \end{aligned}$$

Example 1.107

If a and b are integers such that $\left\| \begin{bmatrix} 2 & \frac{1}{b} \\ 3b & a \end{bmatrix} \begin{bmatrix} b & a \\ \frac{1}{a} & 4 \end{bmatrix} \right\| = 7$, find the sum of all possible values of $a + b$.

Use the $\det(\mathbf{AB}) = \det \mathbf{A} \times \det \mathbf{B}$ to calculate each determinant separately and multiply:

$$\left(2a - 3b \cdot \frac{1}{b}\right) \left(4b - a \cdot \frac{1}{a}\right) = 7$$

Simplify and note that each term on the LHS must be an integer:

$$\underbrace{(2a - 3)}_{\text{Integer}} \underbrace{(4b - 1)}_{\text{Integer}} = (1)(7) = (7)(1) = (-1)(-7) = (-7)(-1)$$

The above four factorizations gives us four cases, each of which we need to check.

Case I:

$$2a - 3 = 1 \Rightarrow a = 2, \quad 4b - 1 = 7 \Rightarrow b = 2 \Rightarrow \text{Valid}$$

Case II:

$$2a - 3 = 7 \Rightarrow a = 5, \quad 4b - 1 = 1 \Rightarrow b = \frac{1}{2} \Rightarrow \text{Not Valid}$$

Case III:

$$2a - 3 = -1 \Rightarrow a = 1, \quad 4b - 1 = -7 \Rightarrow b = -\frac{3}{2} \Rightarrow \text{Not Valid}$$

Case IV:

$$2a - 3 = -7 \Rightarrow a = -2, \quad 4b - 1 = -1 \Rightarrow b = 0 \Rightarrow \text{Not Valid}$$

(where $b = 0$ is not valid because $\frac{1}{b}$ is an element of the matrix)

$$a + b = 2 + 2 = 4$$

1.108: Determinant of a Multiple of a Matrix

$$\det k\mathbf{A} = k^2 \det \mathbf{A}$$

Let

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow k\mathbf{A} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

$$LHS = \det k\mathbf{A} = (ka)(kd) - (kb)(kc) = k^2 ad - k^2 bc = k^2(ad - bc) = k^2 \det \mathbf{A}$$

Example 1.109

The determinant of $\begin{bmatrix} (3 + \sqrt{5})(\sin \theta) & (3 + \sqrt{5})(-\cos \theta) \\ \left(\frac{4}{3 - \sqrt{5}}\right)(\cos \theta) & \left(\frac{4}{3 - \sqrt{5}}\right)(\sin \theta) \end{bmatrix}$ can be written in the form $a + b\sqrt{c}$, where $a, b, c \in \mathbb{Z}$ and c has no perfect square factors. Find $a + b + c$.

Rationalize the terms in the second row:

$$\frac{4}{3 - \sqrt{5}} \times \frac{3 + \sqrt{5}}{3 + \sqrt{5}} = \frac{4(3 + \sqrt{5})}{4} = 3 + \sqrt{5}$$

The matrix simplifies to:

$$\begin{bmatrix} (3 + \sqrt{5})(\sin \theta) & (3 + \sqrt{5})(-\cos \theta) \\ (3 + \sqrt{5})(\cos \theta) & (3 + \sqrt{5})(\sin \theta) \end{bmatrix}$$

Factor $(3 + \sqrt{5})$ out of each term in the matrix:

$$= (3 + \sqrt{5}) \begin{bmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{bmatrix}$$

Calculate the determinant:

$$= \left| (3 + \sqrt{5}) \begin{bmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{bmatrix} \right|$$

Use the property $\det k\mathbf{A} = k^2 \det \mathbf{A}$:

$$= (3 + \sqrt{5})^2 \left| \begin{bmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{bmatrix} \right|$$

Expand using the definition of the determinant:

$$\begin{aligned} &= (3 + \sqrt{5})^2 (\sin^2 \theta + \cos^2 \theta) \\ &= (3 + \sqrt{5})^2 \\ &= 9 + 6\sqrt{5} + 5 \\ &= 14 + 6\sqrt{5} \end{aligned}$$

$$a + b + c = 14 + 6 + 5 = 25$$

B. Minors and Cofactors

1.110: Minor of a Matrix

The minor $M_{i,j}$ of a matrix is the determinant formed by deleting the i^{th} row and the j^{th} column of the matrix.

Example 1.111

Calculate $M_{1,1}$ of

$$\begin{bmatrix} 2 & 8 & 6 \\ 9 & 3 & -3 \\ 4 & -1 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -3 \\ -1 & 6 \end{bmatrix}$$

1.112: Cofactor of a Matrix

The cofactor $C_{i,j}$ is given by:

$$(-1)^{i+j} M_{i,j}$$

- The cofactor differs from the minor only in sign.
- If $i + j$ is even, the cofactor is equal to the minor
- If $i + j$ is odd, the cofactor is the negative of the minor

Example 1.113

$$\begin{bmatrix} 2 & 8 & 6 \\ 9 & 3 & -3 \\ 4 & -1 & 6 \end{bmatrix}$$

- A. Find $M_{2,3}$
 B. Find $C_{2,3}$

$$M_{2,3} = \det \begin{bmatrix} 2 & 8 \\ 4 & -1 \end{bmatrix} = \det \begin{bmatrix} 2 & 8 \\ 4 & -1 \end{bmatrix} = (2)(-1) - (8)(4) = -2 - 32 = -34$$

$$C_{2,3} = (-1)^{2+3} M_{2,3} = -M_{2,3} = 34$$

C. Determinants (3×3)

1.114: Cofactor Expansion

$$\det(\mathbf{A}) = \sum_{j=1}^n (-1)^{i+j} a_{i,j} M_{i,j}$$

Where:

- $a_{i,j}$ is the entry of the i^{th} row and j^{th} column of A
- $M_{i,j}$ is the minor obtained by removing the i^{th} row and the j^{th} column of A .

Example 1.115

Evaluate:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

1.116: 3×3 Expansion

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - afh - bdi - ceg$$

$$\begin{aligned} & a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix} \\ &= a(ei - fh) - b(di - fg) + c(dh - eg) \\ &= aei - afh - bdi + bfg + cdh - ceg \\ &= aei + bfg + cdh - afh - bdi - ceg \end{aligned}$$

1.117: Zero Determinant Property

If a row or a column of a matrix consists entirely of zeroes, then the determinant is zero.

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - afh - bdi - ceg$$

$$\begin{vmatrix} 0 & 0 & 0 \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - afh - bdi - ceg$$

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - afh - bdi - ceg$$

In general, each term has one element from each row, and one element from each column. Hence, if a row or a column is made up of entirely zeroes, each individual term becomes zero and hence the

$$\text{Determinant} = 0$$

1.7 Solving Equations

A. Two Equations in Three Variables

1.118: Two Equations in Three Variables

If we have two equations in three variables:

$$x_1a + x_2b + x_3c = 0$$

$$y_1a + y_2b + y_3c = 0$$

the relation between the variables a , b and c is given by:

$$\frac{a}{\begin{vmatrix} x_2 & x_3 \\ y_2 & y_3 \end{vmatrix}} = \frac{-b}{\begin{vmatrix} x_1 & x_3 \\ y_1 & y_3 \end{vmatrix}} = \frac{c}{\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix}}$$

Multiply the first equation by y_1 , and the second equation by x_1 to make the coefficient of a in both the equations the same:

$$y_1x_1a + y_1x_2b + y_1x_3c = 0$$

$$x_1y_1a + x_1y_2b + x_1y_3c = 0$$

Eliminate a from the system of equations by subtracting the second equation from the first:

$$(y_1x_2 - x_1y_2)b + (y_1x_3 - x_1y_3)c = 0$$

Get one variable on each side on of the equation:

$$(y_1x_2 - x_1y_2)b = -(y_1x_3 - x_1y_3)c$$

Solve for b and c :

$$\frac{-b}{y_1x_3 - x_1y_3} = \frac{c}{y_1x_2 - x_1y_2}$$

Note that the denominator of each fraction is a determinant. But it is the negative of the determinant that we want. Multiply both sides by -1 :

$$\frac{-b}{x_1y_3 - y_1x_3} = \frac{c}{x_1y_2 - y_1x_2}$$

Write each denominator as a determinant, and we get:

$$\frac{b}{\begin{vmatrix} x_1 & x_3 \\ y_1 & y_3 \end{vmatrix}} = \frac{c}{\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix}}$$

Similarly, we can eliminate c from the system of equations to get:

$$\frac{a}{\begin{vmatrix} x_2 & x_3 \\ y_2 & y_3 \end{vmatrix}} = \frac{b}{\begin{vmatrix} x_1 & x_3 \\ y_1 & y_3 \end{vmatrix}}$$

Example 1.119

Find the relation between x , y and z if $x + 2y + 3z = 0$ and $-3x + 2y + 5z = 0$.

Substitute the values into the formula $\frac{a}{\begin{vmatrix} x_2 & x_3 \\ y_2 & y_3 \end{vmatrix}} = \frac{b}{\begin{vmatrix} x_1 & x_3 \\ y_1 & y_3 \end{vmatrix}} = \frac{c}{\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix}}$ to get:

$$\frac{x}{\begin{vmatrix} 2 & 3 \\ 2 & 5 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} 1 & 3 \\ -3 & 5 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 1 & 2 \\ -3 & 2 \end{vmatrix}}$$

Use the formula for a determinant to expand:

$$\frac{x}{10 - 6} = -\frac{y}{5 + 9} = \frac{z}{2 + 6}$$

Simplify:

$$\frac{x}{4} = -\frac{y}{14} = \frac{z}{8}$$

Multiply by 2:

$$\frac{x}{2} = -\frac{y}{7} = \frac{z}{4}$$

B. Writing equations as matrix equations**Example 1.120**

$$2x - 5y + 7z = \frac{1}{2}$$

$$3x + 5y = 5 + 2z$$

$$3x - 2z = 6 - 8y$$

$$\begin{bmatrix} 2 & -5 & 7 \\ 3 & 5 & -2 \\ 3 & 8 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ 5 \\ 6 \end{bmatrix}$$

C. Solving Equations**1.121: Solving Matrix Equations**

$$AX = B$$

Premultiply both sides by A^{-1} :

$$A^{-1}AX = A^{-1}B \Rightarrow IX = A^{-1}B \Rightarrow X = A^{-1}B$$

1.122: Solving Matrix Equations

$$XA = B \Rightarrow X = BA^{-1}$$

Postmultiply both sides by A^{-1} :

$$XAA^{-1} = BA^{-1}$$

$$XI = BA^{-1}$$

$$X = BA^{-1}$$

Example 1.123

$$5x + 7y = 11$$

$$2x - 9y = 4$$

Write the LHS as a matrix, and the RHS as a matrix:

$$\begin{bmatrix} 5x + 7y \\ 2x + 9y \end{bmatrix} = \begin{bmatrix} 11 \\ 4 \end{bmatrix}$$

Note that the LHS can be factored into two matrices:

$$\begin{bmatrix} 5 & 7 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 11 \\ 4 \end{bmatrix}$$

$$\text{Let } \mathbf{A} = \begin{bmatrix} 5 & 7 \\ 2 & 9 \end{bmatrix}, \mathbf{X} = \begin{bmatrix} x \\ y \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 11 \\ 4 \end{bmatrix}$$

$$\mathbf{AX} = \mathbf{B} \Rightarrow \mathbf{X} = \mathbf{A}^{-1}\mathbf{B}$$

$$\text{Substitute } \mathbf{A}^{-1} = \frac{1}{(5)(9)-(7)(2)} \begin{bmatrix} 9 & -7 \\ -2 & 5 \end{bmatrix} = \frac{1}{31} \begin{bmatrix} 9 & -7 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} \frac{9}{31} & -\frac{7}{31} \\ -\frac{2}{31} & \frac{5}{31} \end{bmatrix}:$$

$$\mathbf{X} = \mathbf{A}^{-1}\mathbf{B} = \begin{bmatrix} \frac{9}{31} & -\frac{7}{31} \\ -\frac{2}{31} & \frac{5}{31} \end{bmatrix} \begin{bmatrix} 11 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{99}{31} - \frac{28}{31} \\ -\frac{22}{31} + \frac{20}{31} \end{bmatrix} = \begin{bmatrix} \frac{99}{31} - \frac{28}{31} \\ -\frac{22}{31} + \frac{20}{31} \end{bmatrix} = \begin{bmatrix} \frac{71}{31} \\ -\frac{2}{31} \end{bmatrix}$$

D. Infinite Solutions

Example 1.124

$$\begin{aligned} x + y &= 1 \\ 2x + 2y &= 2 \end{aligned}$$

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\text{Det} \left(\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \right) = (2)(1) - (1)(2) = 2 - 2 = 0$$

1.125: Infinite Solutions

If a system of equations has infinite solutions, then when it is written in matrix form $\mathbf{AX} = \mathbf{B}$, the matrix $\mathbf{A}$ is not invertible.

A linear system of equations in two variables can be represented as a pair of lines.

When the system has infinite solutions, the two lines are the same line, and hence, if we have two equations, one equation is a multiple of the other equation.

$$\begin{aligned} ax + by &= c \\ kax + kby &= kc \end{aligned}$$

$$\begin{bmatrix} a & b \\ ka & kb \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c \\ kc \end{bmatrix}$$

The determinant of matrix $\mathbf{A}$ is

$$a(kb) - b(ka) = kab - kab = 0 \Rightarrow \text{Det} = 0 \Rightarrow \mathbf{A} \text{ is not invertible}$$

E. No Solutions

Example 1.126

$$\begin{aligned}y &= x + 1 \\y &= x + 2\end{aligned}$$

$$\begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

The determinant of $\mathbf{A}$ is:

$$(-1)(1) - (1)(-1) = -1 + 1 = 0$$

F. Review**Example 1.127**

Find the value(s) of θ such that the system of equations below does not have a unique solution:

$$\begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

If the system of equations does not have a unique solution, then the determinant must be zero:

$$\begin{aligned}\cos^2 \theta - \sin^2 \theta &= 0 \\ \cos^2 \theta &= \sin^2 \theta\end{aligned}$$

Take square roots both sides:

$$\cos \theta = \pm \sin \theta$$

The values of the θ that will work are:

$$\frac{\pi}{4} + \frac{\pi}{2}n, n \in \mathbb{Z}$$

Example 1.128

Mark All Correct Options

A system of linear equations $ax + by = c$ and $dx + ey = f$ is written in matrix form:

$$\mathbf{A} = \begin{bmatrix} a & b \\ d & e \end{bmatrix}, \mathbf{X} = \begin{bmatrix} x \\ y \end{bmatrix}, \mathbf{B} = \begin{bmatrix} c \\ f \end{bmatrix} \Rightarrow \mathbf{AX} = \mathbf{B}$$

If $\mathbf{A}$ is not invertible. Then, which of the following can be true:

- A. The system has zero solutions
- B. The system has one solution
- C. The system has two solutions
- D. The system has infinite solutions

Example 1.129

Mark the Correct Option

A system of linear equations $ax + by = c$ and $dx + ey = f$ is written in matrix form:

$$\mathbf{A} = \begin{bmatrix} a & b \\ d & e \end{bmatrix}, \mathbf{X} = \begin{bmatrix} x \\ y \end{bmatrix}, \mathbf{B} = \begin{bmatrix} c \\ f \end{bmatrix} \Rightarrow \mathbf{AX} = \mathbf{B}$$

If $\mathbf{A}$ is not invertible. Then, which of the following must be true:

- A. The system has zero solutions
- B. The system has one solution
- C. The system has infinite solutions
- D. None of the above

1.8 Gauss Jordan Elimination

Example 1.130

$$\begin{bmatrix} 1 & -4 \\ 0 & -10 \end{bmatrix}$$

$$\begin{aligned} x - 4y \\ 0x - 10y \end{aligned}$$

Multiply the second equation by $-\frac{4}{10}$:

Example 1.131

$$\begin{bmatrix} 5 & -20 & 1 \\ -10 & 30 & 1 \end{bmatrix}$$

$$\begin{aligned} \underbrace{5x - 20y = 1}_{\text{Equation I}} \\ \underbrace{-10x + 30y = 1}_{\text{Equation II}} \end{aligned}$$

Multiply Equation I by 2 to get Equation IA:

$$\underbrace{10x - 40y = 2}_{\text{Equation IA}}$$

Add Equation IA to Equation II

$$0x - 10y =$$

$$\begin{aligned} & \begin{bmatrix} 5 & -20 & 1 \\ -10 & 30 & 1 \end{bmatrix} \\ & \text{Multiply Row 1 by 2:} \\ & \begin{bmatrix} 10 & -40 & 2 \\ -10 & 30 & 1 \end{bmatrix} \\ & \text{Add Row 1 to Row 2:} \\ & \begin{bmatrix} 10 & -40 & 2 \\ -10 + 10 & -40 + 30 & \frac{10}{4} - \frac{6}{4} \end{bmatrix} \\ & = \begin{bmatrix} 1 & -4 & 2 \\ 0 & -10 & \frac{3}{4} \end{bmatrix} \\ & \text{Divide } R_2 \text{ by } -10: \\ & = \begin{bmatrix} 1 & -4 & \frac{1}{4} \\ 0 & 1 & \frac{3}{20} \end{bmatrix} \\ & \text{Subtract } 4R_2 = \left(0, 4, \frac{3}{5}\right) \text{ from } R_1: \\ & = \begin{bmatrix} 1 & -4 & \frac{1}{4} \\ 0 & 1 & \frac{3}{20} \end{bmatrix} \end{aligned}$$

1.132: Augmented Matrix

Example 1.133

$$\begin{aligned} -2x + 6y &= 26 \\ 5x - 10y &= -7 \end{aligned}$$

$$\begin{bmatrix} -2 & 6 & 26 \\ 5 & -10 & -7 \end{bmatrix}$$

Divide Row 1 by -2 :

$$\begin{bmatrix} 1 & -3 & -13 \\ 5 & -10 & -7 \end{bmatrix}$$

Multiply Row 1 by 5 to get $[5 \quad -15 \quad -65]$, and subtract it from Row 2:

$$\begin{bmatrix} 1 & -3 & -13 \\ 0 & 5 & 58 \end{bmatrix}$$

Divide Row 2 by 5:

$$\begin{bmatrix} 1 & -3 & -13 \\ 0 & 1 & 11.6 \end{bmatrix}$$

Multiply Row 2 by 3 to get $[0 \quad 3 \quad 34.8]$, and add it to Row 1:

$$\begin{bmatrix} 1 & 0 & 21.8 \\ 0 & 1 & 11.6 \end{bmatrix}$$

Example 1.134

$$\begin{aligned} 2x - y + 4z &= -3 \\ x - 2y - 10z &= -6 \\ 3x + 0y + 4z &= 7 \end{aligned}$$

x	y	z	Constant Terms
2	-1	4	-3
1	-2	-10	-6
3	0	4	7

x	y	z	Constant Terms
1	a	b	c
0	1	d	e
0	0	1	f

$$0x + 0y + z = f \Rightarrow z = f$$

$$0x + y + dz = e$$

$$y + df = e$$

$$y = e - df$$

$$x + ay + bz = c$$

$$x = c - ay - bz = c - a(e - df) + b(f) = c - ae + adf + bf$$

Example 1.135

$$\begin{aligned} 2x - y + 4z &= -3 \\ x - 2y - 10z &= -6 \\ 3x + 0y + 4z &= 7 \end{aligned}$$

Write the augmented matrix:

Write the augmented matrix:

$$\begin{bmatrix} 2 & -1 & 4 & -3 \\ 1 & -2 & -10 & -6 \\ 3 & 0 & 4 & 7 \end{bmatrix}$$

For the first step, we want a 1 in the first row, first column.

One easy way to do it is to interchange Row 1 and Row 2:

$$\begin{bmatrix} 1 & -2 & -10 & -6 \\ 2 & -1 & 4 & -3 \\ 3 & 0 & 4 & 7 \end{bmatrix}$$

Multiply the first row by -2, and add it to the second row.

$$[-2 \quad 4 \quad 20 \quad 12]$$

$$\begin{bmatrix} 1 & -2 & -10 & -6 \\ 0 & 3 & 24 & 9 \\ 3 & 0 & 4 & 7 \end{bmatrix}$$

Multiply the first row by -3, and add it to the third row.

$$[-3 \quad 6 \quad 30 \quad 18]$$

$$\begin{bmatrix} 1 & -2 & -10 & -6 \\ 0 & 3 & 24 & 9 \\ 0 & 6 & 34 & 25 \end{bmatrix}$$

Multiply Row 2 by $\frac{1}{3}$:

$$\begin{bmatrix} 1 & -2 & -10 & -6 \\ 0 & 1 & 8 & 3 \\ 0 & 6 & 34 & 25 \end{bmatrix}$$

Multiply the second row by -6 and add it to the third row:

$$[-6 \quad -6 \quad -48 \quad -18]$$

$$\begin{bmatrix} 1 & -2 & -10 & -6 \\ 0 & 1 & 8 & 3 \\ 0 & 0 & -14 & 7 \end{bmatrix}$$

Multiply Row 3 by $-\frac{1}{14}$

$$\begin{bmatrix} 1 & -2 & -10 & -6 \\ 0 & 1 & 8 & 3 \\ 0 & 0 & 1 & -\frac{1}{2} \end{bmatrix}$$

We have now reached Row Echelon Form.

Multiply Row 3 by 10, and add it to the first Row.

$$[10 \quad 0 \quad 10 \quad -5]$$

$$\begin{bmatrix} 1 & -2 & 0 & -11 \\ 0 & 1 & 8 & 3 \\ 0 & 0 & 1 & -\frac{1}{2} \end{bmatrix}$$

Multiply Row 3 by -8, and add it to the second Row.

$$[0 \quad 0 \quad -8 \quad 4]$$

$$\begin{bmatrix} 1 & -2 & 0 & -11 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -\frac{1}{2} \end{bmatrix}$$

Multiply Row 2 by 2 and add it to the first Row

$$[0 \quad 2 \quad 0 \quad 14]$$

$$\begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -\frac{1}{2} \end{bmatrix}$$

$$(x, y, z) = \left(3, 7, -\frac{1}{2}\right)$$

Example 1.136

Determine when the system of equations below has one solution, no solution, and infinite solutions:

$$\begin{aligned} x + 3y - 3z &= 3 \\ 3x - y + 2z &= 3 \\ 5x + 5y + (a^2 - 13)z &= a + 6 \end{aligned}$$

$$\begin{aligned} 3x + 9y - 9z &= 9 \\ 3x - y + 2z &= 3 \\ 10y - 11z &= 6 \\ 10y &= 6 + 11z \end{aligned}$$

$$\begin{aligned} 5x + 15y - 15z &= 15 \\ 5x + 5y + (a^2 - 13)z &= a + 6 \\ 10y - 15z - (a^2 - 13)z &= 9 - a \\ 10y &= 9 - a + (a^2 + 2)z \end{aligned}$$

$$6 + 11z = 9 - a + (a^2 + 2)z$$

Consider cases to determine the number of solutions.

Case I: (One Solution):

If the variable coefficients are different, then the equation has exactly one solution:

$$\begin{aligned} a^2 + 2 = 11 &\Rightarrow a^2 - 9 = 0 \Rightarrow a = \pm 3 \\ \text{Equation has exactly one solution for } \mathbb{R} - \{\pm 3\} \end{aligned}$$

Case II: The variable coefficients will be same when $a = \pm 3$.

The constants will be same when:

$$6 = 9 - a \Rightarrow a = 3 \Rightarrow \text{Infinite Solution}$$

When $a = -3$, the constant terms will be different, but the variable coefficients will be same.

$$a = -3 \Rightarrow \text{No Solution}$$

2. APPLICATIONS

2.1 Markov Chains (Probability)

A. Probability Basics

2.1: Probability Vector

A n dimensional vector $\vec{r} = (r_1, r_2, \dots, r_n)$ is a probability vector its elements meet the conditions that the sample space of an experiment must satisfy:

$$\begin{aligned} 0 &\leq r_1, r_2, \dots, r_n \leq 1 \\ r_1 + r_2 + \dots + r_n &= 1 \end{aligned}$$

In particular terms, if a probability distribution can be represented as a probability vector.

2.2: Stochastic¹ Matrix

A stochastic matrix is a matrix where the rows, or the columns are probability vectors.

Example 2.3

ABC Cars rents for a day from three locations in New York, called creatively, locations 1, 2 and 3. The matrix on the right shows the locations that a car is rented from in the columns and the locations that a car is returned to in the rows.

	rented from		
returned to	1	2	3
	0.7	0.3	0.3
	0.2	0.2	0.1
	0.1	0.5	0.6

- A. Interpret the meaning of 0.5

The probability that a car rented from location 2 is returned to location 3
= 0.5

A_{ij} is the probability that a car rented from location j is returned to location i .
 i represents the location to which the car is being returned.
 j represents the location from which the car is rented.

2.4: Doubly Stochastic Matrix

If both the rows and the columns of a matrix are probability vectors, then the matrix is doubly stochastic.

B. Short Term Probabilities

2.5: Short Term Probabilities

Given a starting situation at time n , you can calculate the probabilities at $n + 1$

Example 2.6

The Hoyts company has a fleet of 1000 cars for one-day local rentals, which must be leased from either the Exeter office or the Hampton office. The company expects that 70% of the cars rented in Exeter return to the Exeter office, 60% of the cars rented in Hampton return to the Hampton office, and all cars return to one office or the other. Assume that all 1000 cars are rented each day, and that 200 of the rental cars are in Exeter and 800 are in Hampton on Monday morning. What distribution of cars is expected on

- A. Tuesday morning?
B. Wednesday morning? (Phillips Exeter, Math-3)

¹ Stochastic refers to probability.

Let

$$E = \text{Exeter}, \quad H = \text{Hampton}$$

Part A

On Tuesday morning, cars in:

$$\text{Exeter} = E_1 = (0.7) \underbrace{(200)}_{E_0} + (0.4) \underbrace{(800)}_{H_0} = 140 + 320 = 460$$

$$\text{Hampton} = H_1 = (0.3) \underbrace{(200)}_{E_0} + (0.6) \underbrace{(800)}_{H_0} = 60 + 480 = 540$$

		Start	
		E	H
End	E	0.7	0.4
	H	0.3	0.6

$$\begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix} \begin{bmatrix} 200 \\ 800 \end{bmatrix} = \begin{bmatrix} 0.7(200) + (0.4)(800) \\ (0.3)(200) + 0.6(800) \end{bmatrix} = \begin{bmatrix} 460 \\ 540 \end{bmatrix}$$

Part B

On Wednesday morning, cars in:

$$\text{Exeter} = E_2 = (0.7) \underbrace{(460)}_{E_1} + (0.4) \underbrace{(540)}_{H_1} = 322 + 216 = 538$$

$$\text{Hampton} = H_2 = (0.3) \underbrace{(460)}_{E_1} + (0.6) \underbrace{(540)}_{H_1} = 138 + 324 = 462$$

$$\begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix} \begin{bmatrix} 460 \\ 540 \end{bmatrix} =$$

Part C

$$\begin{aligned} E_n &= 0.7E_{n-1} + 0.4H_{n-1} \\ H_n &= 0.3E_{n-1} + 0.6H_{n-1} \end{aligned}$$

Example 2.7: Calculator

Assume students at PEA go to Las Olas or Lexie's once a week. Each week, 51% of the students who went to Lexie's the previous week decide to go to Las Olas and 27% of the students who went to Las Olas the previous week decide to go to Lexie's. Assume that this trend continues indefinitely and that the student population of PEA numbers 1000. If 420 students went to Las Olas last week, how many students should Lexie's expect this week? In two weeks? In three weeks? (Phillips Exeter, Math-3)

Capture the probabilities in a table, and convert the table probabilities into a square matrix:

$$P = \begin{bmatrix} 0.73 & 0.51 \\ 0.27 & 0.49 \end{bmatrix}$$

		Start	
		LO	Lexie's
End	LO	0.73	0.51
	Lexie's	0.27	0.49

Take the distribution of starting students and write it as a column vector:

$$v = \begin{bmatrix} 420 \\ 580 \end{bmatrix}$$

$$\text{Week 1: } P v = \begin{bmatrix} 0.73 & 0.51 \\ 0.27 & 0.49 \end{bmatrix} \begin{bmatrix} 420 \\ 580 \end{bmatrix} = \begin{bmatrix} (0.73)(420) + (0.51)(580) \\ (0.27)(420) + (0.49)(580) \end{bmatrix} = \begin{bmatrix} 602.4 \\ 398.6 \end{bmatrix}$$

$$\text{Week 2: } P^2 v = P(P v) = \begin{bmatrix} 0.73 & 0.51 \\ 0.27 & 0.49 \end{bmatrix} \begin{bmatrix} 602.4 \\ 398.6 \end{bmatrix} = \begin{bmatrix} (0.73)(602.4) + (0.51)(398.6) \\ (0.27)(602.4) + (0.49)(398.6) \end{bmatrix} = \begin{bmatrix} 642.528 \\ 357.472 \end{bmatrix}$$

$$\text{Week 3: } P^3 v = P(P^2 v) = \begin{bmatrix} 0.73 & 0.51 \\ 0.27 & 0.49 \end{bmatrix} \begin{bmatrix} 642.528 \\ 357.472 \end{bmatrix} = \begin{bmatrix} (0.73)(642.528) + (0.51)(357.472) \\ (0.27)(642.528) + (0.49)(357.472) \end{bmatrix} =$$

C. Steady State: Algebra

2.8: Long Term Probabilities/ Steady State Vector

The long-term probabilities can be calculating using a vector that does not change with the transition matrix.

$$Pw = w$$

Where

$$P = \text{Matrix of Transition probabilities}$$

$$w = \text{long term state}$$

Example 2.9: Calculator

Let's assume students at PEA go to Las Olas or Lexie's once a week. Each week, 51% of the students who went to Lexie's the previous week decide to go to Las Olas and 27% of the students who went to Las Olas the previous week decide to go to Lexie's. Assume that this trend continues indefinitely and that the student population of PEA numbers 1000. If 420 students went to Las Olas last week, how many students should Lexie's expect this week? In two weeks? In three weeks? In the long run? (Phillips Exeter, Math-3)

Let the number of students who go to

$$\text{Las Olas} = x \Rightarrow \text{Lexie's} = 1000 - x \Rightarrow w = \begin{bmatrix} x \\ 1000 - x \end{bmatrix}$$

In the long run, the distribution of the past week is equal to the distribution of the current week. Hence, we must solve for w such that

$$Pw = w$$

$$\begin{bmatrix} 0.73 & 0.51 \\ 0.27 & 0.49 \end{bmatrix} \begin{bmatrix} x \\ 1000 - x \end{bmatrix} = \begin{bmatrix} x \\ 1000 - x \end{bmatrix}$$

Carry the multiplication using the definition:

$$\begin{bmatrix} 0.73x + 0.51(1000 - x) \\ 0.27x + 0.49(1000 - x) \end{bmatrix} = \begin{bmatrix} x \\ 1000 - x \end{bmatrix}$$

$$\begin{bmatrix} 0.22x + 510 \\ -0.22x + 490 \end{bmatrix} = \begin{bmatrix} x \\ 1000 - x \end{bmatrix}$$

For two matrices to be equal, their components must be equal.

Equate the components to get a system of equations:

$$0.22x + 510 = x \Rightarrow x = \frac{510}{0.78} = 653.846$$

$$\begin{aligned} -0.22x + 490 &= 1000 - x \\ 0.78x &= 510 \end{aligned}$$

D. Steady State: Matrices

/

2.10: Solving for the steady state

$$(I - P)w = 0$$

$$Pw = w$$

Subtract w from both sides:

$$Pw - w = 0$$

Factor w on the LHS:

$$(P - I)w = 0$$

$$(I - P)w = 0$$

Example 2.11

$$(I - P)w = 0$$

$$\left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} .8 & .3 & .2 \\ .1 & .2 & .6 \\ .1 & .5 & .2 \end{bmatrix} \right) \begin{bmatrix} x \\ y \\ 1-x-y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\frac{1}{10} \begin{bmatrix} 2 & -3 & -2 \\ -1 & 8 & -6 \\ -1 & -5 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1-x-y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} \text{1st Row: } 2x - 3y - 2(1 - x - y) &= 0 \Rightarrow y = 4x - 2 \\ \text{2nd Row: } -x + 8y - 6(1 - x - y) &= 0 \Rightarrow 5x + 14y = 6 \end{aligned}$$

Substitute value of y from 1st Row in 2nd Row:

$$\begin{aligned} 5x + 14(4x - 2) &= 6 \Rightarrow x = \frac{34}{61} \\ y &= 4x - 2 = 4\left(\frac{34}{61}\right) - 2 = \frac{136}{61} - \frac{122}{61} = \frac{14}{61} \\ 1 - x - y &= 1 - (x + y) = 1 - \left(\frac{34}{61} + \frac{14}{61}\right) = 1 - \frac{48}{61} = \frac{13}{61} \end{aligned}$$

$$w = \begin{bmatrix} \frac{34}{61} \\ \frac{14}{61} \\ \frac{13}{61} \end{bmatrix}$$

Rented from Location

1	2	3	
.8	.3	.2	1
.1	.2	.6	2
.1	.5	.2	3

Returned to Location

E. Further Theory

2.12: Transient and Recurring State

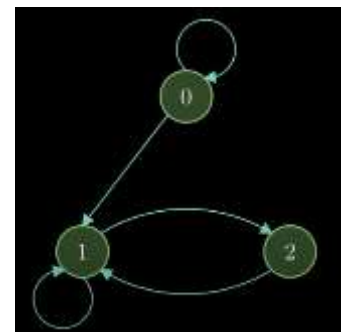
If there is a possibility of leaving the state and never returning, that state is called a transient state.

A Markov chain comes back to a recurrent state, infinitely often, with probability 1.

Example 2.13

In the diagram, 0 is a transient state.

In the diagram, 1 and 2 are recurrent states.



2.14: Irreducible and Reducible Chain

If you can go from one state to any other state, then it is an irreducible chain.

A reducible chain has restrictions where some states cannot be reached from other states. It can be divided into irreducible chains.

2.15: Conditions for Stationary State

We found stationary state using the equation

$$(I - P)w = 0$$

For the stationary state to exist, the conditions are:

- Markov Chain must be irreducible
- Markov chain must be aperiodic

F. References

[Playlist 2.16](#)

On Markov Chains

2.2 Transformations

A. Basics

2.17: Scaling

For a constant k , the matrix $\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$, $k > 0$ defines a scaling.

- $k > 1$ is a stretch
- $0 < k < 1$ is a shrink

Example 2.18

Consider the point $(1,3)$, and the matrix $\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$. Find the product of the point and the matrix.

$$\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \times 1 + 0 \times 3 \\ 0 \times 1 + \frac{1}{2} \times 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{3}{2} \end{bmatrix}$$

2.19: Rotation

The matrix of a counterclockwise rotation in $\mathbb{R}^2$ through an angle θ is given by

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

This matrix has the form:

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

Example 2.20

Consider

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Show that

- A. it is of the form $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$
- B. $a^2 + b^2 = 1$
- C. The column vectors are unit vectors

$$a^2 + b^2 = \cos^2 \theta + \sin^2 \theta = 1$$

First Column Vector is $\mathbf{v} = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$

$$|\mathbf{v}| = \sqrt{x^2 + y^2} = \sqrt{\cos^2 \theta + \sin^2 \theta} = \sqrt{1} = 1$$

Challenge 2.21

Use a rotation matrix to prove the identity:

$$\cos(a + b) = \cos a \cos b - \sin a \sin b$$

$$\sin(a + b) = \sin a \cos b + \cos a \sin b$$

$$\cos(a - b) = \cos a \cos b + \sin a \sin b$$

$$R(a) = \begin{bmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{bmatrix}$$

$$R(b) = \begin{bmatrix} \cos b & -\sin b \\ \sin b & \cos b \end{bmatrix}$$

$$R(a + b) = \begin{bmatrix} \cos(a + b) & -\sin(a + b) \\ \sin(a + b) & \cos(a + b) \end{bmatrix}$$

But recall that rotations are commutative:

$$[R(a)][R(b)] = [R(a + b)]$$

$$\begin{bmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{bmatrix} \begin{bmatrix} \cos b & -\sin b \\ \sin b & \cos b \end{bmatrix} = \begin{bmatrix} \cos(a + b) & -\sin(a + b) \\ \sin(a + b) & \cos(a + b) \end{bmatrix}$$

$$\begin{bmatrix} \cos a \cos b - \sin a \sin b & -(\sin a \cos b + \cos a \sin b) \\ \sin a \cos b + \cos a \sin b & \cos a \cos b - \sin a \sin b \end{bmatrix}$$

$$\cos(a - b) = \cos a \cos b + \sin a \sin b$$

Use $\cos(a + b) = \cos a \cos b - \sin a \sin b$

$$\cos(a - b) = \cos a \cos(-b) - \sin a \sin(-b)$$

Use $\cos(-\theta) = \cos \theta, \sin(-\theta) = -\sin \theta$

$$\cos a \cos b - (-\sin a \sin b)$$

$$\cos a \cos b + \sin a \sin b$$

2.22: Reflection

The matrix

$$\begin{bmatrix} a & b \\ b & -a \end{bmatrix}, \quad a^2 + b^2 = 1$$

Is a reflection

Example 2.23

Consider the matrix

$$X = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{bmatrix}$$

- A. Show that the column vectors of this matrix are unit vectors.
 B. Find the product of the point (1,2) and the matrix above.

$$v = \begin{bmatrix} \frac{3}{5} \\ -\frac{4}{5} \end{bmatrix} \Rightarrow |v| = \sqrt{\left(-\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \sqrt{\frac{9}{25} + \frac{16}{25}} = \sqrt{\frac{25}{25}} = \sqrt{1} = 1$$

$$\begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -\frac{3}{5} \times 1 + \frac{4}{5} \times 2 \\ \frac{4}{5} \times 1 + \frac{3}{5} \times 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -\frac{3}{5}a + \frac{4}{5}b \\ \frac{4}{5}a + \frac{3}{5}b \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$-\frac{3}{5}a + \frac{4}{5}b = a \Rightarrow \frac{4}{5}b = \frac{8}{5}a \Rightarrow 4b = 8a \Rightarrow b = 2a$$

$$\frac{4}{5}a + \frac{3}{5}b = b$$

2.3 Review and Challenge

A. Review

Example 2.24

Consider a system of equations

$$\begin{aligned} ax + by &= c \\ dx + ey &= f \end{aligned}$$

which is represented using matrices as

$$\begin{bmatrix} a & b \\ d & e \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c \\ f \end{bmatrix}$$

Let $A = \begin{bmatrix} a & b \\ d & e \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$, $B = \begin{bmatrix} c \\ f \end{bmatrix}$

If A is invertible, then we have:

$$AX = B \Rightarrow A^{-1}AX = A^{-1}B \Rightarrow IX = A^{-1}B \Rightarrow X = A^{-1}B$$

However, if A is not invertible then, what can we conclude about the system of equation given above?

25 Examples